higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

ELECTROTECHNICS N5

(8080085)

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09:00–12:00

This question paper consists of 5 pages and a formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
ELECTROTECHNICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Write neatly and legibly.

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QUESTION 1

1.1 State THREE methods of improving commutations. (3)

1.2 An eight-pole DC motor has a lap-wound armature with 820 conductors. The armature current is 648 A. Assuming that the ratio of the armature ampere turns per pole to the interpole ampere turns per pole is 0,88.

Determine the number of turns per pole that is required for the interpole (5)

1.3 Determine the resistance values of the first three elements of a starter for a 480 V DC shunt motor if the starter is to have fifteen studs and the starting current is not to exceed 1,5 A. The resistance of the armature circuit is 0,35 ohms. (7)

1.4 A 440 V, 120 kW shunt generator has 1 350 turns on each pole of its field winding. On a no-load a current of 2,5 A in the field winding, produces a terminal voltage of 440 V, while on full load the shunt current has to be increased to 4,5 A for the same terminal voltage and the same speed.

Calculate the number of series field turns required per pole for level compounding. (5)

[20]

QUESTION 2

2.1 A circuit consisting of a coil having an inductance of 185 micro-henry and a resistance of 8 ohms is connected in parallel with a variable capacitor. This combination is then connected in series with a resistor of 6 580 ohms. This circuit is connected across a 440 V supply with a frequency of 1,4 MHz.

Calculate the following:

2.1.1 The capacitance of the capacitor required to give resonance (3)

2.1.2 The impedance of the parallel circuit (2)

2.1.3 The current in each branch of the parallel circuit (5)

2.2 A constant voltage at a frequency of 0,95 MHz is applied across a circuit consisting of an inductor in series with a variable capacitor. When the capacitor is set to 287 pF, the current has its maximum value. When the capacitance is reduced to 271 pF, the current is 0,74 of its maximum value.

Calculate the following:

2.2.1 The inductance of the inductor (3)

2.2.2 The resistance of the inductor (7)

[20]

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QUESTION 3

3.1 Briefly explain how iron losses are affected by a change of the frequency of a transformer. (2)

3.2 When does a transformer have a negative voltage regulation? (1)

3.3 A 125 kVA, 2 400/600 volt, single-phase transformer, operating at no-load has a percentage resistance of 3 % and a percentage leakage reactance of 4,2%. The primary supply voltage is 2 400 volts and 0,86 power factor lagging.

Determine the secondary terminal voltage at full load. (11)

3.4 A 13 kVA, 2 100/200 V single-phase transformer has a percentage impedance of 2 + j4 %.

Calculate the following referred to the secondary side of the transformer:

3.4.1 The equivalent resistance (4)

3.4.2 The equivalent reactance (2)

[20]

QUESTION 4

4.1 A 240 V three-phase motor has a full-load output of 20 kW and the power factor is 0,89. The efficiency at full load is 88 per cent.

Calculate the reading of the two wattmeters used to measure the input power. (8)

4.2 A three-phase overhead transmission line has a resistance of 4 ohms per phase and an inductance of 25,465 mH per phase. The line delivers a 12 MW star-connected load at a power factor of 0,8 lagging. The voltage between lines at the receiving end is 25 V and the frequency is 50 Hz.

Calculate the following:

4.2.1 The phase voltage at the sending end (9)

4.2.2 The voltage regulation in percentage (2)

4.3 The load taken by a three-phase induction motor was measured by the two-wattmeter method and the readings were 820 W and 300 W.

Calculate the total power of this induction motor. (1)

[20]

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QUESTION 5

5.1 How does a high voltage affect the performance of an induction motor? (2)

5.2 Explain why an induction motor cannot develop torque when running at a synchronous speed. (4)

5.3 A three-phase star-connected alternator, driven at 1 500 r/min is required to generate a line voltage of 500 volts at 50 Hz in an open circuit. The stator has 6 slots per pole per phase and 10 conductors per slot. The distribution factor is 0,97. Assume full-pitched coils.

Calculate the following:

5.3.1 The number of poles (2)

5.3.2 The useful flux per pole (4)

5.4 A three-phase, 50 Hz, 6-pole induction motor has a slip of 6% when the output is 45 kW. The frictional loss is 285 W.

Calculate the following:

5.4.1 Rotor speed (2)

5.4.2 Rotor copper loss (6)

[20]

TOTAL: 100

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ELECTROTECHNICS N5

FORMULA SHEET

Armature ampere-turns/pole
Ankerampèrewindings/pool

$$= \frac{1}{2} \cdot \frac{I_a}{C} \cdot \frac{Z}{2P}$$

$$E = V \pm I_a R_a$$

$$E = \frac{2pNZ\Phi}{60c}$$

$$T = 0{,}318 \ \frac{I_a}{c} \ ZP\Phi$$

$$k = n\sqrt{\frac{R_1}{r_m}}$$

$$r_1 = R_1 \left[\frac{k-1}{k}\right]$$

$$r_1 = R_s \ \frac{1-y}{1-y^m}$$

$$R_1 = bR_1 \ (k-1) \times \frac{1-b^n}{1-b} + r_m$$

$$y = \frac{I_2}{I_1}$$

$$r_1 = bR_1 \ (k-1)$$

$$\frac{E_1}{E_2} = \frac{K\Phi_1 N_1}{K\Phi_2 N_2}$$

$$\frac{T_1}{T_2} = \frac{K\Phi_1 I_{a1}}{K\Phi_2 I_{a2}}$$

$$I_{ave/\,gem} = \frac{i_1 + i_2 + i_3 + \ldots i_n}{n}$$

$$I_{rms/\,wgk} = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + \ldots + i_n^2}{n}}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$f = \frac{1}{2\pi L}\sqrt{\frac{L}{C} - R^2}$$

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$$P = \sqrt{3} I_L V_L \, Cos \, \phi$$

$$P_1 = V_L I_L \, Cos \, (30 - \phi)$$

$$P_2 = V_L I_L \, Cos \, (30 + \phi)$$

$$\tan \phi = \frac{\sqrt{3} \, (P_2 - P_1)}{(P_2 + P_1)}$$

% Voltage regulation
% Spanningsreëling

$$= I_1 \frac{(R_e \, Cos\phi \pm X_e \, Sin\phi)}{v_1} \times \frac{100}{1}$$

$$Z_e = \sqrt{R_e^2 + X_e^2}$$

$$\%Z_e = \frac{I \, Z_e}{V} \times \frac{100}{1}$$

$$S_1 = S \frac{Z_2}{Z_1 + Z_2}$$

$$E = 2{,}222 \; k_d k_p \, Z \Phi f$$

$$I_r = \frac{E_r}{Z_r}$$

$$E_o = V_p \frac{Z_r}{Z_s}$$

$$Cos\phi_r = \frac{R}{Z_r}$$

$$s = \frac{2\pi T (n_s - n_r)}{2\pi T \, n_s}$$

$$L = 0{,}05 + 0{,}2 \, Lin \, \frac{d}{r}$$

$$C = \frac{1}{36 Lin \, \frac{d - r}{r}}$$

$$C = \frac{1}{18 Lin \, \frac{de}{r}}$$

% Regulation
% Regulering

$$= \frac{V_s - V_R}{V_R} \times \frac{100}{1}$$

$\frac{V_S}{V}$

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